Davies-Bouldin Index Project

Background

In MVO2.0, we use K-Means to preselect stocks. However, we need to manually set the random seed number R for K-Means, making it not deterministic. So I want to develop a strategy to select an optimal R for K-Means to solve the problem.

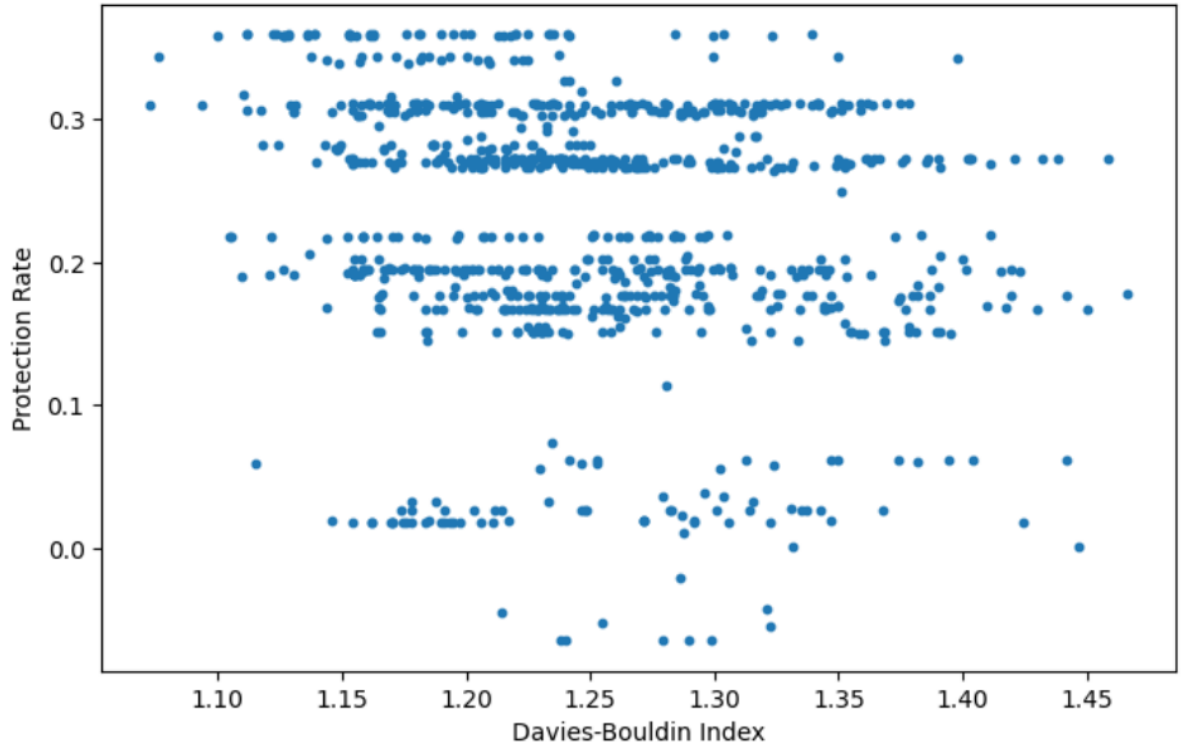
Davies-Bouldin Index

The Davies–Bouldin index (DBI) is a metric for evaluating clustering algorithms. This is an internal evaluation scheme, where the validation of how well the clustering has been done is made using quantities and features inherent to the dataset. The metric considers distances both between clusters and within each cluster.

The idea of the project is to select the seed number R that gives the best Davies-Bouldin score and use that R for K-Means. And we are hoping that a better separated clustering gives a better result for MVO.

Experiment 1:

Run K-Means for R=1~900 and record the davies-bouldin index and protection rate for each R. Put them in a scatter plot, as shown below.



Experiment 2:

Run K-Means for R=1~900 and take the top trials. Compare the preselected stocks given by these trials and combine them to give a new set of preselected stocks.

A. Take all the stocks from the top 3 trials and pass them to K-Means to generate a new cluster with a new centroid. Protection Rate=38.67%
B. Take all the stocks from the top 3 trials and calculate average beta for each stock (use ‘average’ method). Protection Rate=36.37%
C. Take all the stocks from the top 5 trials and calculate average beta for each stock (use ‘average’ method). Protection Rate=34.26%